DUSTREAM EARNED VALUE MANAGEMENT (EVM) ANALYSIS

Activity	Predecessor	Duration (Weeks)	Cost $ (Per Week)	Budget Cost (BC)
A	-	1	$8,000	$8,000
B	A	2	$5,000	$10,000
C	A	3	$6,800	$20,400
D	B	3	$4,500	$13,500
E	B	5	$5,500	$27,500
F	A	10	$8,700	$87,000
G	C	10	$7,300	$73,000
H	D	5	$5,300	$26,500
I	B	10	$11,400	$114,000
J	E	5	$9,700	$48,500
K	D	5	$4,400	$22,000
L	A	10	$9,200	$92,000
M	E	5	$3,500	$17,500
			BAC	$559900

Activity	*Use cell colour fill in the table cells provided below to build your Gantt Chart (Precedence Diagram Method) for this project. It will help if you colour the duration in dark blue and the completed in red)*														
A															
B															
C															
D															
E															
F															
G															
H															
I															
J															
K															
L															
M															
Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

The progress report at the beginning of Week 9 (as shown by the red line in the Gantt Chart above), gives the information provided in the left-hand table below. From this information perform the calculations listed in the table provided below and to the right (round to 2 decimal places for CPI/SPI and to the nearest dollar for other figures). Once you have done this, you need to write a short progress report to the Project Sponsor-listing and explaining the issues and making recommendations. The report template is on the second page of this document.

Progress Report Start of Week 9		
Activity	% Complete	Incurred Cost
A	100.0%	$9,500
B	100.0%	$8,900
C	66.6%	$13,800
D	66.6%	$9,100
E	80.0%	$21,900
F	90.0%	$76,300
G	10.0%	$7,800
H	20.0%	$5,000
I	70.0%	$84,000
J	40.0%	$19,500
K	20.0%	$9,000
L	90.0%	$82,000
M	20.0%	$2,500

				Cost Implications		Schedule Implications	
Activity	AC (ACWP)	EV (BCWP)	PV (BCWS)	CPI	CV	SPI	SV
A	$9,500	$8,000	$8,000	0.84	-$1500	1	$0
B	$8,900	$10,000	$10,000	1.12	$1100	1	$0
C	$13,800	$13586.4	$20,400	0.98	-$213.6	0.67	-$6813.6
D	$9,100	$8991	$13,500	0.99	-$109	0.67	-$4509
E	$21,900	$22000	$27,500	1	$100	0.8	-$5500
F	$76,300	$78300	$60900	1.03	$2000	1.29	$17400
G	$7,800	$7300	$29200	0.94	-$500	0.25	-$21900
H	$5,000	$5300	$10600	1.06	$300	0.5	-$5300
I	$84,000	$79800	$57000	0.95	-$4200	1.4	$22800
J	$19,500	$19400	$0	0.99	-$100	--	$19400
K	$9,000	$4400	$8800	0.49	-$4600	0.5	-$4400
L	$82,000	$82800	$64400	1.01	$800	1.29	$18400
M	$2,500	$3500	$0	1.4	$1000	--	$3500
Total	$349300	$343377.4	$310300	0.98	-$5922.6	1.11	$33077.4

Note: Red indicates over budget/ behind the schedule

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Provide the following analysis information

What are the implications in relation to cost (e.g. are we over budget, under, budget, etc.)? Explain this in terms of CPI & CV. Additionally, make sure that you explain the implications in plain English, so the reader will be able to understand what this means.	The project is slightly over budget in terms of work completed. The CPI is 0.98 and CV is -$5922.6. CPI value 0.98 refers that for every $1 spent, the project is only receiving approximately $0.98 worth of budgeted work. The negative cost variance means that the project has spent approximately $5,923 more than the budgeted value of the work completed. The main cost concerns are Activities I and K, which have highest negative cost variances of $4,200 and $4,600 respectively. Activity A also has a negative cost variance of $1,500. These issues should be investigated to determine whether they result from inefficient resource use, unexpected expenses or inaccurate cost estimates. Although the overall cost overrun is relatively small compared with the total project budget, corrective action is required to prevent further deterioration.
What are the implications in relation to schedule (e.g. are we ahead or schedule or behind schedule)? Explain this in terms of SPI & SV and make sure that you link back to a discussion of what is shown in the PDM diagram you created on Page 1 of this report. In referring to the timeline, discuss whether individual activities/tasks are ahead of, or behind, time and how progress on various activities may be affecting the project overall (e.g. are delays in completing some tasks holding back the completion of dependent activities – hint look at this in terms of predecessors).	The SPI is approximately 1.11 and the SV is positive $33,077.4. This indicates that, based on the supplied planned value figures, the project has earned approximately 11% more value than was planned by the beginning of Week 9. However, this overall result does not mean that every activity is progressing on schedule. Activities C, D, E, G, H and K have negative schedule variances, indicating that they are behind their planned progress. Activity G has the largest negative schedule variance at $21,900, followed by Activity C at $6,813.6. These delays are important because the PDM diagram shows dependencies between activities. Activity G depends on Activity C, while Activities H and K depend on Activity D. Therefore, delays in Activities C and D may affect the timely completion of their dependent activities. Activity E is also behind schedule and is a predecessor of Activities J and M. Delays in E could consequently affect the start or completion of both tasks. In contrast, Activities F and L show positive SVs, while Activity I also appears ahead of schedule based on its supplied figures. The project manager should assess whether resources can be redeployed from activities with schedule flexibility to delayed activities without creating new bottlenecks. Overall, despite the positive aggregate SPI, the project manager must monitor individual activities and their dependencies carefully. The planned value figures should also be checked against the original Gantt chart to ensure the overall schedule assessment is accurate.

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What is the Budget at Completion (BAC)? (Note: This is the sum of the Budget Cost (BC) as shown in the first table on Page 1).	The Budget at Completion (BAC) is $559900. This represents the total approved budget for completing all activities in the project, assuming the original budget estimates remain unchanged.
What is your Estimate at Completion (EAC)?	Estimate at Completion (EAC) = BAC / CPI = $559900/0.98 = $571326.53 This forecast assumes that the current cost performance continues throughout the remainder of the project. Since the CPI is below 1, the project is expected to cost more than its original budget if current spending efficiency does not improve. The forecast suggests a final cost exceeding the approved budget, so the project manager should investigate the causes of the cost overruns and implement corrective measures.
What is the Variance at Completion (VAC), and what are the implications of this result?	Variance at Completion (VAC) = BAC - EAC = $559900-$571326.53 = -$11426.53 The negative value indicates that the project is forecast to exceed its approved budget by approximately $11426.53 if current cost performance continues. The project manager should review contractor expenditure, monitor remaining work closely and seek to improve resource efficiency. Any proposed budget changes should be discussed with the Project Sponsor and approved through the appropriate change-control process.
What is the level of effort required to complete the tasks, according to TCPI in relation to BAC? Express this as a TCPI numeric and explain the implications in terms of the calculated value. Please remember that the TCPI on its own may sometimes be somewhat misleading. Therefore, also think about the level of effort required in terms of progress (e.g. think about the implications in relation to the SPI & SV results).	TCPI = (BAC − EV) / (BAC − AC) = $216522.6 / $210600 = 1.03 The TCPI in relation to BAC is approximately 1.03. This means that, to complete the project within the original budget, the team must achieve approximately $1.03 worth of budgeted work for every $1 of remaining expenditure. This is a higher level of cost efficiency than the current CPI of 0.98, so an improvement in cost performance is required. However, TCPI only measures the financial efficiency required to meet the budget. It does not account for the practical difficulties of completing delayed activities on time. Although the overall SPI calculated from the supplied figures is approximately 1.11, several individual activities are behind schedule, including Activities C, D, E, G, H and K. The project manager must therefore improve both cost efficiency and task progress. Resources should be redirected towards delayed activities where possible, while ensuring that activities with dependencies can proceed without unnecessary waiting time.

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What recommendations for managing this situation would you provide to the Project Sponsor (PS)? Think about this realistically. To help you understand the implications for our contractors, the separation of tasks is as follows: DCPlus is responsible for Activities A, C, D, E, G, I, K, L and M, whereas DemSet is responsible for completing Activities B, F, H, and J. Remember, you will need to identify how you would get the contractors to complete their tasks on time. In particular, make sure you consider how you would get them to redeploy resources (if possible) from Activities that are ahead of time to help groups complete other tasks (e.g. applying smoothing/levelling). In this scenario, you can assume that all members of the two contractors can do any of the required tasks.

- DemSet should review resource availability across Activities B, F, H and J. Activity F has a positive schedule variance and CPI above 1, suggesting that some resources may be available for reassignment, subject to its remaining workload and criticality. These resources could assist with delayed activities such as H, which is also managed by DemSet. DCPlus should similarly review Activities I and L, which have positive schedule variances and determine whether spare resources can be reassigned to Activities C, D, E, G or K
- DCPlus should focus on Activities C and D because they are predecessors to Activities G, H and K respectively. Activity E should also receive attention because it is a predecessor to Activities J and M. Improving progress on these predecessor activities will reduce the risk of further delays to dependent tasks
- DCPlus should investigate the poor cost performance of Activities A, C, I and K, while DemSet should review expenditure on its activities, particularly where costs are high relative to completed work. The project manager should examine labour hours, productivity, rework and any unplanned expenses. Additional resources should only be assigned where their expected benefit justifies the cost
- Both contractors should coordinate their schedules to avoid leaving resources underutilised on activities that are ahead of schedule while other tasks are falling behind. Resource smoothing should be used where work can be rescheduled within available float without changing the project completion date. Resource levelling or direct reassignment should be considered where resource constraints are causing delays, with the impact on the project completion date carefully assessed. Since all contractor staff can perform any required task, cross-allocation of suitably available staff is a practical option
- Both contractors should provide weekly updates showing actual expenditure, percentage complete, forecast completion dates and any issues affecting dependent tasks. The project manager should update the Gantt chart and EVM measures regularly and escalate significant cost overruns or schedule risks to the Project Sponsor